

CSE360- Computer Interfacing
Quiz 04; Spring'26
Set A

Name: _____ **ID:** _____ **Section:** _____

Marks: 15 points

Time: 20 minutes

Instructions:

Question 2 (CO2 - 8 points): Suppose, you are designing a security system for a shopping mall. Now the first requirement is to check if any customer is carrying any metal object.

- a) **Explain** the working mechanism of another sensor that can detect only metal objects without any physical contact. **State** your reasons on why the sensor cannot detect anything else other than metal.
- b) Next, you want to include a few smoke sensors for safety. **Name** the two types of Smoke sensors. **Express** the working mechanism of both types in detail.
- c) Suppose you'd also like to add a Toxic Gas Sensor to the system. **Discuss** the working principle of a Gas Sensor.
- d) Name a sensor that uses piezoelectric material to sense environmental data.
- e) Classify all the sensors above a passive or active.